Stock Movement Prediction from Price Trends and News Sentiment

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Abstract

Within recent years, machine learning has emerged as one of the most powerful tools in predicting stock performances for investors. Such machine learning algorithms integrate market data, financial reports, and macroeconomic data to predict stock movement and volatility; however leading methods systematically overreact to macroeconomic news or fail to take it into account completely. A central aim of our project is to quantify how exchange-traded fund (ETF) relationships reflect underlying macroeconomic conditions, and how these relationships influence individual stock behavior. To address this limitation, we constructed a data management process linking historical company-specific news headlines to corresponding stock performance data across 60 individual tickers.
(Add 1-2 more sentences for method once done with ML side) (Add 2-3 more sentences on results)

Keywords: Machine Learning, Exchange-Traded Funds(ETF's), Batch Gradient Descent, Ordinary Least Squares

1 Introduction

2 Methods

To train our model, we followed a conventional machine learning development model.

2.1 Data Sourcing and Preprocessing

2.1.1 Dataset Identification

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2.1.1 Dataset Merging

3 Results

4 Discussion

5 Conclusion

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Author Contributions

- **Honore Alexander**
- **Owen Holt**
- **Pranavi Khanna**
- **Dylan Lim**: Created application and backend for model inference, designed the initial frontend. Other contributions to source research and technical report documentation.
- **Yihong Li**: Researched sources and drafted most of the technical report sections. Set up LaTeX and formatted the report. Set up and designed frontend for model inference.
- **Ethan Lee**
- **Dan Firstenberg**
- **Hyeongseung Nam**
- **Oscar Pineda**: Researched sources for the report, contributed to writing the report
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